Kalamata Sai Manideep

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EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore *Aug 2023 – July 2027*
B.Tech in Computer Science with specialization in Artificial Intelligence. **CGPA:** 6.5
Tirumala junior college *June 2021 – May 2023*
Board of Intermediate Education. **Percentage:** 96.6%
Sri chaitanya techno curriculum *June 2020 – May 2021*
Board of Secondary Education. **Percentage:** 99.33%

SUMMARY

I am a B.Tech Computer Science student specializing in Artificial Intelligence, with strong skills in programming, machine learning, deep learning, and IoT systems. I have built end-to-end solutions including air-quality monitoring with disease prediction, real-time CTG analysis using hybrid DL models, and other applied AI applications. My work spans sensor integration, model development, backend systems, and user-friendly interfaces, reflecting a solid ability to deliver practical, data-driven solutions. I am committed to continuous learning and applying AI technologies to solve impactful real-world problems.

SKILLS

OS	Windows	**Databases**	SQL
Languages	C, Python, C++	**Web**	HTML, CSS, JavaScript
Frameworks	TensorFlow, PyTorch, Flask	**Coursework**	Data Structures,Dbms, ML

PROJECTS

Brain Tumor Prediction Using MRI Scans

- Built an automated MRI analysis pipeline using median filtering, histogram equalization, and Otsu's thresholding for clean tumor segmentation.
- Extracted shape, texture (GLCM), and intensity features using contour detection and convex hull algorithms.
- Trained a Random Forest classifier to predict tumor type (glioma, meningioma, pituitary, no-tumor) with interpretable, low-compute processing.
- **Tools Used:**
  Python (ML/DL), OpenCV (image preprocessing, contour detection), NumPy SciPy (matrix operations, filtering) , scikit-image (texture/thresholding), scikit-image (GLCM texture features, thresholding), scikit-learn (Random Forest classifier, evaluation)

Real-Time Prediction of Delivery Type and Cesarean Risk Using FHR AND TOCO Signals

- Developed a hybrid deep learning model combining Conv1D–LSTM with statistical feature encoding to analyze fetal heart rate (FHR) and uterine contraction (UC) signals.
- Implemented a dual feature engineering pipeline, extracting 18 global statistical features and sequential 60-sec window features for robust modeling.
- Designed a multi-task learning architecture to simultaneously predict delivery type, Cesarean risk, pH, and pCO values.
- TOOLS USED : Python, TensorFlow/Keras, NumPy, SciPy, Pandas, Matplotlib, CTU-UHB CTG Dataset

IoT-Based Air Quality Monitoring and Disease Prediction Using deep Learning

- Built a real-time IoT system using ESP32 integrated with MQ-7, MQ-135, Dust Sensor (GP2Y1010AU0F), and DHT11 to measure NO, SO, PM2.5, PM10, dust, temperature, and humidity.
- Implemented a data pipeline to transmit sensor readings to a Flask backend via Wi-Fi in JSON format for processing and prediction.
- Trained a Feed-Forward Neural Network (FFNN) to classify diseases based on environmental parameters, achieving high accuracy across multiple metrics (MSE, MAE, R^2).

Hostel meal feedback system

- Developed a cloud-based web application for managing hostel meal feedback with secure role-based authentication (Students Admins).
- Designed a normalized relational database schema (Users, Students, Admins, Meals, Feedback) ensuring clean data flow, integrity, and efficient querying
- Implemented meal scheduling, feedback submission, rating storage, and suggestion tracking using structured SQL tables.
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- **Tools Used:** Microsoft Azure (App Service, SQL Database), SQL, HTML/CSS/JavaScript, Node.js/Express, Git GitHub.